FLOATING SOLAR PROJECT OF GREENAM ENERGY

STEERING THE PLANET TO NET ZERO

Document Prepared by EKI Energy Services Limited

Project Title	Floating Solar project of Greenam Energy
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Floating Solar project of Greenam Energy (henceforth called "the project") is a greenfield renewable energy power generation project (floating solar PV plant), supplying electricity to the captive consumer. The total capacity of the project is 24.7MWp (22 MWac). The project is developed on the surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district, Tamil Nadu, India.

The project is implemented by SPIC and Greenstar fertilizers via Greenam Energy Private Limited (GEPL) a SPV made for this project and energy generated is supplied to Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL), via a grid interface through a wheeling for captive usage. A common power purchase agreement (PPA) had been executed between SPIC and GFL and GEPL.

Further the GEPL is jointly owned by SPIC Ltd (SOUTHERN PETROCHEMICAL INDUSTRIES CORPORATTION LIMITED), Greenstar Fertilizers Limited (GFL) and AM International Holding Private Limited (AMIH) the same can be identified as certificate of ownership as per rule 3 of Electricity Rule 2005. Greenam Energy Private limited is a SPV stablished by SPIC, Greenstar and AMIH for this purpose of Captive generation. Hence the project activity is meeting the criteria as mentioned in AM0123: Renewable energy generation for captive use.

The purpose of this Project is to utilize the area of the reservoir to generate clean energy for exporting to use by the captive consumer, which in absence of project activity would have been supplied by existing Indian grid, which is based mostly on coal and oil-based fossil fuels.

The project boundary includes the project power plant (project site where the solar power plant has been installed including the solar power plant, power evacuation infrastructure, energy metering points, switch yards and other civil constructions) implemented in the state and all other power plants/units connected physically to the national (Indian) grid that the project power plants is connected to.

The average annual generation of the project is 41,241 MWh for the current crediting period. The emission reduction from the project is estimated to be around average 38,376 tCO_2e /annum and a cumulative emission reduction of 268,629 tCO_2e for the entire crediting period of 7 years.

There was no activity at the site of the project proponent prior to the implementation of this Project activity. The project participant was not involved in generation of any renewals power and supplying it to the grid under the pre-project scenario, therefore, in the absence of the Project activity the baseline scenario would be either continued use of fossil fuel dominated grid connected electricity generation or fossil fuel used in power plant or carbon intensive electricity.

Audit Type	Period	Program	VVB Name	Number of years
Validation	12/01/2022 - 11/01/2029[1]	VCS	LGAI Technological Center S.A. (Applus+ Certification)	7 (renewable)
Verification	12-January-2022 to 28-February-2023	VCS	LGAI Technological Center S.A. (Applus+ Certification)	1 Years, 1 Months, 17 Days
Total	**12-January-2022 to 28-February-2023**	**VCS**	**LGAI Technological Center S.A. (Applus+ Certification)**	**1 Years, 1 Months, 17 Days**

1.2 Sectoral Scope and Project Type

The project is falling into the sectoral scope 1 (Energy renewable/non-renewable sources),

The project is not a grouped project.

CDM approved consolidated Methodology - AM0123: Renewable energy generation for captive use --- Version 1.0[2]

1.3 Project Eligibility

Project activity is Floating solar power generation in India. Floating solar is not in Excluded category for Non LDC. In line with the Section 2.1.1 of VCS Standard V4.5 the project eligibility is assessed below:

It results in CO_2 emission reductions, one of the seven Kyoto Protocol greenhouse gases.

As per section 2.1.3 of VCS Standard ver-4.5 table no. 1 excluded project activities-

(The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems e.g., batteries).

The project activity is floating solar; thus the project activity is eligible under the scope of the VCS program.

[1] Date of final validation report

[2] https://cdm.unfccc.int/methodologies/DB/F0UCO5LLYB1LDTVD8GCTJZKACTGUTQ

1.4 Project Design

The project has been designed to include a single location or installation only, It is not being developed as a grouped project.

Eligibility Criteria

Not Applicable

1.5 Project Proponent

Organization name	SOUTHERN PETROCHEMICAL INDUSTRIES CORPORATTION LIMITED
Contact person	P.Senthil Nayagam
Title	Director
Address	"SPIC HOUSE", 88 Mount Road, Guindy, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912

Organization name	Greenstar fertilizers limited
Contact person	P.Senthil Nayagam
Title	Director
Address	"SPIC HOUSE", 88 Mount Road, Guindy, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

Organization name	Greenam Energy Private limited
Contact person	P.Senthil Nayagam
Title	Director
Address	"SPIC HOUSE", 88 Mount Road, Guindy, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912

Email	senthilnayagam@spic.co.in

Organization name	AM International Holding Private Limited
Contact person	P.Senthil Nayagam
Title	Director
Address	"SPIC HOUSE", 88 Mount Road, Guindy, Chennai, Tamil Nadu, India, 600032
Telephone	+91 9443311912
Email	senthilnayagam@spic.co.in

1.6 Other Entities Involved in the Project

Organization name	EKI Energy Services Ltd.
Role in the project	Project Representative
Contact person	Mr. Manish Dabkara
Title	MD & CEO
Address	Office No 201, Plot No 48, Scheme 78, Part 2, Vijay Nagar, Indore-452010, Madhya Pradesh
Telephone	+91 9589899649
Email	registry@enkingint.org, Sandeep.kurmi@enkingint.org

1.7 Ownership

Project is implemented by SPV, Greenam Energy, which is owned by Southern Petrochemical Industries Corporation Limited (SPIC), and Greenstar fertilizers (GFL) and AM International Holding Private Limited[3], the same can be verified from Ownership document shared by SPIC as per rule 3 of Electricity Rule 2005.

AM International Holding Private Limited Owns all the three i.e Greenam[4], SPIC and GFL.The commissioning certificate for project activity are the supporting documents to demonstrate the

[3] https://aminternational.sg/businesses/fertilizers-supply-chain/

[4] https://aminternational.sg/businesses/green-solutions/

project ownership. This demonstrates the right of use according to clause 3.7.1 (3) of VCS Standard (v4.5 – "a project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals". Also, other legal compliances may be considered;

- Commissioning certificate
- Wheeling agreement
- Purchase orders and O&M Agreement

1.8 Project Start Date

The start date of the project activity is 12-January-2022. which is the date of commissioning of the project.

1.9 Project Crediting Period

Total Crediting period is 3*7 Years (renewable)

First crediting period is from 12-January-2022 to 11-January-2029

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
Year 1 (12-January-2022 to 31-December-2022)	37,614
Year 2 (01-January-2023 to 31-December-2023)	38,726
Year 3 (01-January-2024 to 31-December-2024)	38,559

Year 4 (01-January-2025 to 31-December-2025)	38,394
Year 5 (01-January-2026 to 31-December-2026)	38,229
Year 6 (01-January-2027 to 31-December-2027)	38,064
Year 7.a from (01-January-2028 to 31-December-2028)	37,900
Year 7.b from (01-January-2029 to 11-January-2029)	1,142
Total estimated ERs	268,629
Total number of crediting years	7
Average annual ERs	38,376

1.11 Description of the Project Activity

The 24.7MWp (22 MWac) floating solar PV plant is developed on the surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district in the Tamila Nadu, India. Project is installed in 64 acres of water body surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district, Tamil Nadu, India.

The power generated by the floating solar PV plant have evacuated to the utility substation at 110kV. Depending on the quantum of power injected into the substation as recorded by the custody transfer meter, the state utility i.e., TANGEDCO would allocate power to the two off takers based on prior agreements with GEPL. Wheeling charges shall be borne by the off takers.

The project activity is the installation of a new 22 MW AC Solar grid connected power plant/unit at the site where no renewable power plant was operating prior to the implementation of the project activity (green-field plant)

Details of project activity solar power project is outlined in the table below:

Project Proponent	Capacity of the Power plant	Date of Commissioning	State	Connectivity
Greenam Energy Pvt Ltd	22 MW(AC)	12/01/2022[5]	Tamil Nadu	Indian Grid (SPIC and GFL)

The electricity generated from project activity (operational solar power plant) is connected to the state grid. The project activity thus reduces the anthropogenic emissions of greenhouse gases

The Project Activity is a new facility (Greenfield) and the electricity generated by the project will be exported to identified user through Indian electricity grid. The Project Activity will therefore displace an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid. The standard estimated lifetime of the project activity is considered as 25 years[6] for solar technology. This may increase depending on the operation & maintenance of the plant. In the Pre- project scenario the entire electricity, delivered by the grid for use by the AMIH Group companies - Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL), would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

Table: 1.11(a) Technical Specification:

Components	Make
Modules (free issue)	Jinko solar
Inverters	Sungrow Inverters
Transformer	Power transformer – Hammond Power Solution Inverter Transformer – Essennar
VCB	ABB
MMS	Ceil & Terre
Module Cleaning	Considered manual cleaning with reservoir water
Weather Monitoring station and Sensors	Kipp and Zonen
SCADA	Trinity Touch
Remote Monitoring System (RMS)	Trinity Touch

Table: 1.11(b) Transformer details

Current Req No	1. 1 No of Transformer (20-22 MVA) 2. 4 No's of Transformer (6.250 MVA * 3 No's, 3.250 MVA*1 No)
Future Expansion	NA
Details	Power Transformer 1. 20/22 MVA *1 No

[5] Date of Grid synchronization of the project 22MW is 12/01/2022

[6] https://cercind.gov.in/2020/regulation/159_reg.pdf

	2. Ratio – 11 KV / 110 KV 3. Cooling – ONAN/ONAF 4. Winding – Copper 5. HV Connection – Bushing; LV Connection – Cable Box 6. HV Winding - Star, LV winding – Star 7. Vector Group – YNYN0 8. Impedance -10% Max Inverter Transformer 1. 6250MVA & 3250MVA – 4 No's 2. Ratio – 0.600 KV / 11 KV 3. Cooling – ONAN 4. Winding – Copper 5. HV Connection – Cable box; LV Connection – Cable Box 6. HV Winding - Delta, LV winding – Star 7. Vector Group – Dy11y11/ Dy11

Modules numbers may change in case of increase in module power rating.

Table: 1.11(c) Modules detail

Modules Details	1. 1500 V System 2. Make –Jinko Solar , Type: Mono PERC, Rated capacity -385 Wp 3. Qty – 65800 no's

Table: 1.11(d) Modules detail

Inverters Details	1. 1500 V System 2. Make –Sungrow , Type -Central Inverter 3. 3125 KVA@50deg, 1500 Vdc/600 Vac 4. Qty – 7no's

(GHGs) associated with equivalent amount of electricity generation from the existing grid connected power plants (mostly fossil fuel based) and from addition of new generation sources into the grid. The average annual generation of the project is 41,776 MWh and after considering degradation factor average annual generation will be 41,241 MWh. The emission reduction from the project is estimated to be around 38,376 tCO_2e /annum and a cumulative emission reduction of 268,629 tCO_2e for the entire crediting period of 7 years

1.12 Project Location

Greenam Energy Private Limited (22 MW) project activity is located in Muthiahpuram village Thoothukudi district in Tamil Nadu, with the geographical Coordinates of 8°44'6"N to 78°7'44"E

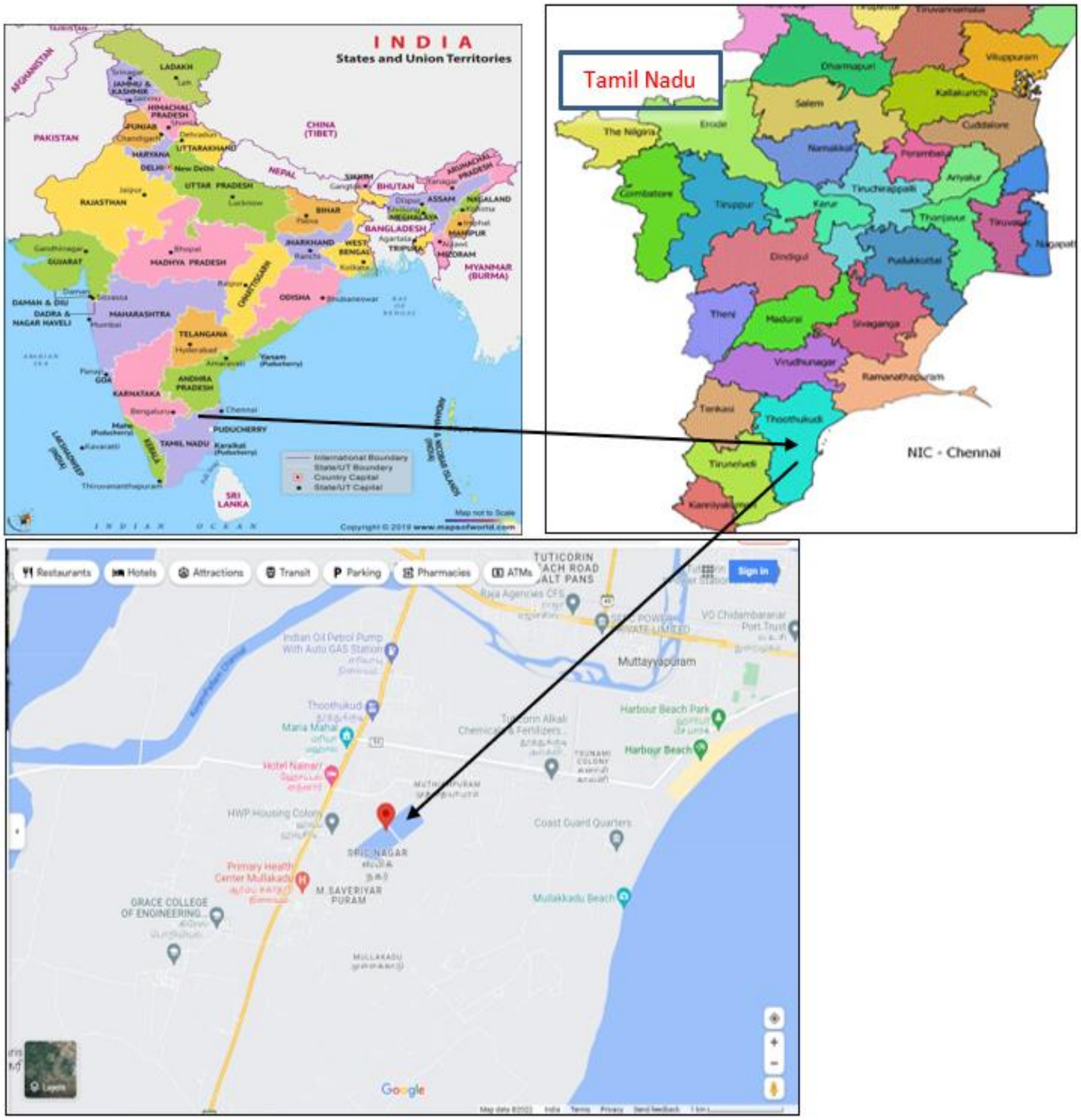

Figure 1: Location of the Project

1.13 Conditions Prior to Project Initiation

There was no project at the site of the project participant prior to the implementation of this project activity. The project participant was not involved in generation of solar PV based power and supplying it to the grid under the pre-project scenario, therefore, in the absence of the project activity, the equivalent amount of electricity would have been generated from the connected / new power plants in the INDIAN GRID, while project activity is a green field project. Therein, the main emission source in the pre-project scenario is the fossil fuel-based power plants connected to the UNIFIED INDIAN GRID[7] and the primary GHG involved is CO_2

[7] https://cea.nic.in/cdm-co2-baseline-database/?lang=en

Prior to the initiation of the project activity, the equivalent amount of electricity would have been drawn from grid connected or new power plants, in national Grid. The grid is predominantly coal based and therefore is a major source of carbon dioxide emissions in India. The main emission in the pre project scenario is the power plants connected to the national Grid, and main GHG emission is CO_2.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Indian relevant laws and regulations. This project Is voluntary initiatives by the Project Proponent.

- Electricity Act, 2003[8] and its subsequent amendment as it does not influence the choice of fuel used for power generation nor has it mandated power generation using renewable energy only.
- Environmental (Protection) Act, 1986[9] and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006[10] and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)[11]
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003[12].
- Noise Pollution (Regulation and Control) Amendment Rules, 2017[13]
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016[14] and amended in November, 2021.
- E-waste (Management and Handling) Rules, 2020[15].

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[8] https://cercind.gov.in/Act-with-amendment.pdf

[9] https://parivesh.nic.in/writereaddata/ENV/eprotect_act_1986.pdf

[10] http://www.environmentwb.gov.in/pdf/EIA%20Notification,%202006.pdf

[11] https://cpcb.nic.in/air-pollution/#:~:text=The%20Air%20(Prevention%20and%20Control,of%20air%20pollution%20in%20India.

[12] https://cpcb.nic.in/water-pollution/#:~:text=The%20Water%20(Prevention%20and%20Control%20of%20Pollution)%20Cess%20Act%20was,certain%20types%20of%20industrial%20activities.

[13] https://upload.indiacode.nic.in/showfile?actid=AC_CEN_16_18_00011_198629_1517807327582&type=rule&filename=2555%20dated%2010.08.2017%20noise%20pollution%20.pdf

[14] https://cpcb.nic.in/rules/

[15] https://cpcb.nic.in/e-waste/

- Plastics Waste Management Rules, 2016 amended in 2021[16].
- Batteries (management and handling) rules 2020[17].

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not participated in any other GHG Programs. This project initially applied for for registration in other GHG program in absence of eligibility in VCS. Now, It has been cancel out (paused) from the GCC and applied for the VCS program for which declaration on double accounting has been provided. It is declared that PP shall not apply registration of the same project activity in other GHG mechanism. PP shall submit withdrawal letter from other mechanism once the project is registered in GCS.

1.15.2 Projects Rejected by Other GHG Programs

The Project is not rejected in any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits.

1.16.2 Other Forms of Environmental Credit

The Project has no intent to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

1.16.3 Supply Chain (Scope 3) Emissions

Not Applicable, as the project is not a supply chain project.

[16] https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1745433#:~:text=and%20Climate%20Change-,Government%20notifies%20the%20Plastic%20Waste%20Management%20Amendment%20Rules%2C%202021%2C%20prohibiting,from%20the%2031st%20December%2C%202022.

[17] https://www.eqmagpro.com/wp-content/uploads/2020/02/Battery-Waste-Management-Rules-2020-draft.pdf

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project achieves significant GHG emission reductions by harnessing the power from the Floating solar project. The project activity includes 22 MWac floating solar PV plant on the surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district, Tamil Nadu, India. Thus, the project also provides long-term job opportunities to local people and contributes to the economic prosperity. Which directly contributes to the achieving following SDG 8. Whereas project also contributes in achieving SDG 7 and SDG 13 goals.

1.17.2 Sustainable Development Contributions Activity Monitoring

The project activity fulfils SDG Target 7.2 By 2030, increase substantially the share of renewable energy in the global energy mix, 8.5 By 2020, Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities and 13.2 Integrate climate change measures into national policies, strategies and planning of SDG Goals.

- SDG 7.0: According to SDG target 7.0, indicator 7.2.1, the project activity generates electricity from renewals energy sources that ensures supply of electricity sources from clean energy sources at affordable rate. This project activity will supply 288,687 MWh clean electricity for entire crediting period.

- SDG 8.0: According to SDG target 8.0, indicator 8.5.1, Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities the project activity creates employment opportunity for 10 individuals for entire crediting period.

- SDG 13.0: According to SDG target 13.0, indicator 13.2.2, it states “Take urgent action to combat climate change and its impact”. By mitigating 2,68,629 tCO_2e which is greenhouse gas, the project activity has contributed towards SDG target 13.0. for entire crediting period.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	The implemented activities to increase	The project activity includes the installation of 22 MW of renewable energy capacity that will deliver zero-emission electricity annually. The project has also contributed to the SDG goals by supplying 38,856 MWh of renewable electricity to the grid	The project is likely to Contribute 38,856 MWh renewable energy share in total grid energy for the current monitoring period.
2)	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	The Implemented activities to increase	The project has contributed to the SDG goals by creating employment opportunities for 02 persons.	Project will also upgrade employee's knowledge and capacity over and above the commonly used technology in the energy sector of India. And likely creation of employment opportunity for 05 individuals for current monitoring period.
4)	13.0	13.2.2 Tonnes of greenhouse gas emissions avoided or removed	The Implemented activities to increase	The project activity through implementation of 22 MW of floating Solar power generation unit has result in reduction of 36,174 tCO_2e for current monitoring period.	The project activity will result in the reduction of the total 36,174 tCO_2e throughout the Monitoring period.

1.18 Additional Information Relevant to the Project

Leakage Management

Not appliable as the project is not a AFOLU project.

Commercially Sensitive Information

No commercial sensitive information has been excluded from the public version of the project description.

Further Information

There are no information or incidents that will have bearing on the eligibility of the project, the net GHG emission reductions or removals, or the quantification of the project 's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

There was no harm identified form the project and hence no mitigations measures are applicable.

2.2 Local Stakeholder Consultation

Local stakeholders were invited by sending invitation letter on 02-March-2022 to the President of SoosaiNagar. The consultation was carried out on 11-March-2022 within plant premises. Total 22 nos. of Stakeholders consulted in the meeting.

The local stakeholder consultation process was carried out through a physical meeting at the project site. As a part of the consultation process, the project representatives explained all stakeholders about the type of development through the project and the technology used for power generation towards enhancing stakeholders' awareness about the project. During the meeting the project representative talked about the employment opportunities and other benefits likely to be rendered through this project and its contribution towards fighting against climate change.

The project Proponent also shared a feedback form during individual interviews during focused group discussions where the stakeholder can put down his/her complaint and the same if found genuine were to be addressed immediately. Also, regular stakeholder engagement is one the key focus at the site.

The project stakeholders were totally in support for setting up of these kinds of projects in the region.

Q: Will there be free supply of power to the local people?

A: The generated power will be fed in the grid. Project promoter can't supply directly power to the local people. They have to get authorized connection from Govt. body. But due to the project activity the supply of power in the area will increase.

Q: Will there be employment generation due to the project activity for youth from the adjoining areas?

A: Responding about the increased possibilities for employment of local youth due to the project activity, it was pointed out that preference would be given for locals in the employment opportunities.

Q: Will it impact the underground water level in the nearby area?

A: No, it will not impact the underground water level of the nearby area.

As a part of continuous feedback from stakeholders, the grievances register is being placed at site and is being continuously monitored and addressed through the grievances cell on regular basis and maintained in a register at site office. Stakeholders can provide their inputs or concerns over the project anytime. The site manager is responsible for addressing the grievances received from the stakeholders. The same was also informed during the stakeholder consultation meeting. Additionally, PP will consult stakeholders annually to check the requirements of any developmental activities in the villages and PP may allot the CSR fund to activities based on the need basis. The mechanism will ensure to continue to engage with local stakeholders in an ongoing proactive manner.

The list of the Attendees is attached in the appendix 1.

2.3 Environmental Impact

The guidelines on Environmental Impact Assessment have been published by Ministry of Environment, Forests, and climate change (MoEFCC), Government of India (GOI) under Environmental Impact Assessment notification 14-September-2006[18]. Further amendments to the notification have been done on 14-July-2018. As per the notification:

"The following projects or activities shall require prior environmental clearance from the concerned regulatory authority, which shall hereinafter referred to be as the Central Government in the Ministry of Environment and Forests for matters falling under Category 'A' in the Schedule and at State level the State Environment Impact Assessment Authority (SEIAA) for matters falling

[18] http://www.environmentwb.gov.in/pdf/EIA%20Notification,%202006.pdf

under Category 'B' in the said Schedule, before any construction work, or preparation of land by the project management except for securing the land, is started on the project or activity:

I. All new projects or activities listed in the Schedule to this notification;

II. Expansion and modernization of existing projects or activities listed in the Schedule to this notification with addition of capacity beyond the limits specified for the concerned sector, that is, projects or activities which cross the threshold limits given in the Schedule, after expansion or modernization;

III. Any change in product – mix in an existing manufacturing unit included in Schedule beyond the specified range."

As the Solar power generation projects are not listed in any of the categories of the schedule, it does not require Environmental Impact Assessment.

The environmental impacts of the project activity are not considered to be significant by the project participant or the host party.

2.4 Public Comments

The project activity was published for public comment and no any relevant comments were received during the one month commenting period from 24-March-2023 to 23-April-2023.

2.5 AFOLU-Specific Safeguards

Not Applicable

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Methodology Title and reference: AM0123 - Renewable energy generation for captive use (Version 01.0)[19]

The applied methodology draws upon the following tools:

Tool 02 - Combined tool to identify the baseline scenario and demonstrate additionality (Version 07.0 EB 93 Annex 3)[20]

[19] https://cdm.unfccc.int/UserManagement/FileStorage/CDM_ACMN1RJXZGUELGP70ULR0CKZHM13BMD18

[20] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-02-v7.0.pdf

Tool 07 - Tool to calculate the emission factor for an electricity system (Version 07.0, EB 100, Annex 4)[21]

Tool 24 – Common practice (Version 03.1, EB 84, Annex 7)[22]

Tool 27- Investment analysis (Version 12.0, EB 116 annex 2)[23]

Tool 3: “Tool to calculate project or leakage CO2 emissions from fossil fuel combustion version 3.0[24]

Tool 5: “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, TOOL05, Version 03.0[25]

3.2 Applicability of Methodology

The choice of the AM0123 methodology is accurate since the proposed project activity respects all the applicability conditions required. The applicability of the methodology is established through assessment of the following conditions as provided in below table. The applicability condition 6 & 7 as per AM0123 (Ver 01.0) is not discussed below as the project is Solar PV Power plant, whereas both the above conditions are relevant to hydro power plants, hence not applicable to project scenario.

Para	Applicability Conditions, AM0123 (Ver 01.0)	Justification of eligibility
2	This methodology applies to a greenfield renewable energy power generation project activities supplying electricity to the captive consumer. Where necessary the greenfield power plant may be integrated with battery energy storage systems (BESS).	VCS project activity is a greenfield renewable energy power generation project (floating solar PV plant), supplying electricity to the captive consumer. Hence, the project activity meets the applicability condition of the methodology.
3	This methodology is applicable to renewable energy power generation project activities that install a greenfield power plant supplying electricity to the captive consumer via: 1. A grid interface through a wheeling; or 2. A dedicated electricity transmission and distribution line.	Energy generated by VCS project activity shall be supplied to Captive users via a grid interface through a wheeling. Hence, the project activity meets the applicability condition of the methodology.

21 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

22 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf

23 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf

24 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-03-v3.pdf

25 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf

4	In case the project activity integrates a BESS, the methodology is applicable only to project activities that integrate greenfield BESS with a greenfield renewable energy power plant.	Project activity is greenfield power project and not integrated with BESS, hence para 4 is not relevant to the project activity.
5	The methodology is applicable under the following conditions: (a) In the pre-project scenario, the captive consumer does not source electricity from a renewable energy source; (b) The renewable energy producer and captive consumption facility shall be owned by the same project participant; (c) The project activity may include a renewable energy power plant/unit of one of the following types: solar power plant/unit, wind power plant/unit, and/or hydro power plant/unit with or without a reservoir. (d) The greenfield project power plant should, at the time of the start date of the project activity, be contractually bound to supply electricity to a captive consumer at least for the entire duration of the first crediting period. (e) The project renewable energy plant(s) shall not export more than 10 per cent on a yearly basis of its generation to a grid. This condition shall be met over the crediting period; (f) In case of integration of BESS as per paragraph 4 above, the project participants shall demonstrate that the BESS was an integral part of the design of the project activity (e.g., by referring to feasibility studies or investment decision documents); (g) The BESS should be charged with electricity generated from the project renewable energy power plant(s). Only during exigencies	(a) In the pre-project scenario, the captive consumer does not source electricity from a renewable energy source; Hence, the project activity meets the applicability condition of the methodology. (b) Greenam Energy Private Limited (GEPL), Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL) are group companies of AM International Holdings Pte. Limited. Hence, it can be concluded that the renewable energy producer (GEPL) and captive consumption facility (SPIC & GFL) are owned by the same project participant. Hence, the project activity meets the applicability condition of the methodology. (c) The project activity involves installation of the "greenfield power project" generating electricity using solar plant. Hence, the project activity meets the applicability condition of the methodology. (d) Project activity is contractually bound before implementation via a long term common power purchase agreement (PPA) executed between GEPL and the off-takers, namely SPIC and GFL. Hence, the project activity meets the applicability condition of the methodology. (e) Project activity supplies 100% of electricity generated to captive consumer and this condition will be verified for the entire crediting period. Hence, the project

	may the BESS be charged with electricity from the grid or a backup generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.3 below. The charging using the grid or backup generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant on an annual basis. During the time periods (e.g., days(s), week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant, for the concerned periods of the monitoring period, shall consider baseline emissions as zero, however, they shall account the project emissions as per requirements under section 5.4.3. Below.	activity meets the applicability condition of the methodology. (f) Project activity is greenfield power project and not integrated with BESS, hence this condition is not relevant to the project activity. (g) Project activity is greenfield power project and not integrated with BESS, hence this condition is not relevant to the project activity.
8	In addition, the applicability conditions included in the tools referred to below apply.	Applicability conditions of the tools included has been justified below.

Applicability conditions of Tool 2: “Combined tool to identify the baseline scenario and demonstrate additionality (version 07)

Applicability Conditions	**Justification of eligibility**
Tool 02: Combined tool to identify the baseline scenario and demonstrate additionality (version 07.0)	
The tool is applicable to all types of proposed project activities. However, in some cases, methodologies referring to this tool may require adjustments or additional explanations as per the guidance in the respective methodologies. This could include, inter alia, a listing of relevant alternative scenarios that should be considered in Step 1, any relevant types of barriers other than those presented in this tool and guidance on how common practice should be established.	Applied methodology AM0123 para 11(a) refers to Tool 2 hence this tool is applicable to the project activity.

Applicability conditions of Tool 3: “Tool to calculate project or leakage CO2 emissions from fossil fuel combustion version 3.0

Applicability Conditions	Justification of eligibility
Tool 03: Tool to calculate project or leakage CO2 emissions from fossil fuel combustion	
This tool provides procedures to calculate project and/or leakage CO2 emissions from the combustion of fossil fuels. It can be used in cases where CO2 emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify to which combustion process j this tool is being applied.	Applied methodology AM0123 para 42 says that No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel, Hence this tool is not applicable to the project activity.

Applicability conditions of Tool 5: "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, TOOL05, Version 03.0

Applicability Conditions	Justification of eligibility
Tool 5: "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, TOOL05, Version 03.	
1. If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid.The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) offgrid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to	The project activity instances do not have a non-renewable component

Applicability Conditions	Justification of eligibility
the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/ electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Project activity instances supplies electricity to consumers and hence the tool is applicable.
3. This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO2 emissions.	There are no captive renewable power generation technologies are installed to provide electricity in the baseline scenario, Hence not applicable

Applicability conditions of Tool 7: "Tool to calculate the emission factor for an prove electricity system", v.7.0

Applicability Conditions	Justification of eligibility
Tool 07: Tool to calculate the emission factor for an electricity system (Version 07.0)	
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This condition of applicable, OM, BM and CM are estimated using this tool (under section B.6.1) for calculating of the baseline emission.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option II.a and option IIb. If option II.a is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of	Since the project activity is grid connected, the condition is applicable and emission factor has been calculated accordingly.

Applicability Conditions	Justification of eligibility
the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in India, a Non-Annex I country. Therefore, this condition is not applicable to the project activity.
Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	The project activity is a green field solar power plant and hence the condition of biofuel emission factor is not applicable.

Applicability conditions of Tool 24: “Common practice (Version 03.1)

Applicability Conditions	Justification of eligibility
Tool 24: Common practice (Version 03.1)	
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality	The project activity is applying the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality
In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in	The applied approved baseline and monitoring methodology dose not define approaches for the

Applicability Conditions	Justification of eligibility
this methodological tool, the requirements contained in the methodology shall prevail.	conduction of the common practice test.

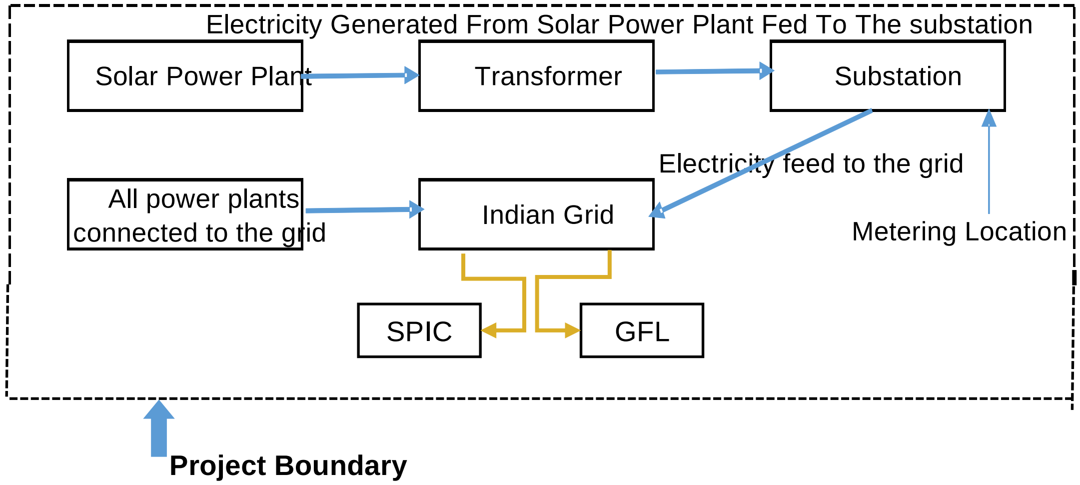
Applicability Conditions	Justification of eligibility
Tool 27: Investment analysis (Version 12.0)	
Para 2 - This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario. – Proposed project activity is “Tool for the demonstration and assessment of additionality”	The project activity applies Tool for the demonstration and assessment of additionality.
Para 3 - In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail. – Methodology refers “General guidelines for SSC CDM methodologies, information on additionality (attachment A to Appendix B)	The applied approved baseline and monitoring methodology contains requirements for the investment analysis that are not different from those described in this methodological tool.

This comparison shows clearly that version 1.0 of AM0123 is applicable to the proposed project activity.

3.3 Project Boundary

As per para 18 of AM0123 (Ver 01.0), *The spatial extent of the project boundary includes the project power plant/unit and or all power plants/units connected physically to the electricity system that the project power plant is connected to, the captive consumer that receives electricity generated by the project activity via wheeling, or through a dedicated electricity transmission and distribution line.*

Hence, the spatial extent of the project boundary includes the project power plant / unit and all power plants / units connected physically to the electricity system that the project power plant is connected to. The project boundary therefore includes the project site where the power plant has been installed, associated power evacuation infrastructure, energy metering points, switch yards the national grid of India.



The project boundary includes CO_2e as greenhouse gas (GHG) that would have emanated from the operation of grid connected fossil fuel-based power plants in the absence of the project activity.

Source		Gas	Included?	Justification/Explanation
Baseline	CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to project activity	CO_2	Yes	Main emission source
		CH_4	No	Minor emission source
		N_2O	No	Not Applicable
Project	Solar energy projects under the project activity	CO_2	No	As a zero-emission grid connected Solar power project no emissions will result from the project activity in operation phase.
		CH_4	No	As zero-emission grid connected
		N_2O	No	Solar power project no emissions will result from the project activity.

3.4 Baseline Scenario

As per para 21 of AM0123 (Ver 01.0), baseline scenario would be defined as per step 1 of Tool 2:

Furthermore, as explained in Section 3.5 of the VCS PD-MR, two alternatives are selected alternative S1: *"No investment is undertaken by the project participant to meet the electricity demand of the captive consumer, and electricity delivered to the captive consumer would have* *been generated by the operation of grid-connected power plants and by the addition of new generation sources".*

Alternative S4: The project participant makes an investment to meet the electricity demand of a captive consumer through the construction of captive power plant using renewable power generation technologies, i.e., the proposed project activity is undertaken without being registered as a CDM project activity;

Alternative 4 is not plausible scenario as project is not financially viable option and need revenue from carbon credit. Financial analysis is explained in section 3.5 of this report.

Hence Alternative 1 has been selected as the appropriate baseline alternative for this project activity, which is in line with the selected methodology. Further as per para 22 of AM0123, scenario 1 is economical most attractive alternative scenario. This is proved by using investment analysis in section 3.5.

3.5 Additionality

The additionality of the Project shall be demonstrated by applying the following approach, consisting of two components:

i. A Legal Requirement Test; and

ii. An Additionality Test either based on a Positive List test or a projects-specific additionality test.

A Legal Requirement test:

The project is not mandated/enforced by law and is entirely a voluntary activity. Since voluntary commitments/agreements within a sector or by an entity do not constitute the legal requirement.

Further the project confirms its commissioning and operation is not mandated by any law and as per below rules:

- Power generation using renewable energy is not a legal requirement or a mandatory option[26].
- There are state and sectoral policies, framed primarily to encourage solar power projects.
- These policies have also been drafted realizing the extent of risks involved in the projects and to attract private investments.
- The Indian Electricity Act, 2003[27] (May 2007 Amendment) does not influence the choice of fuel used for power generation.
- There is no legal requirement on the choice of a particular technology for power generation.

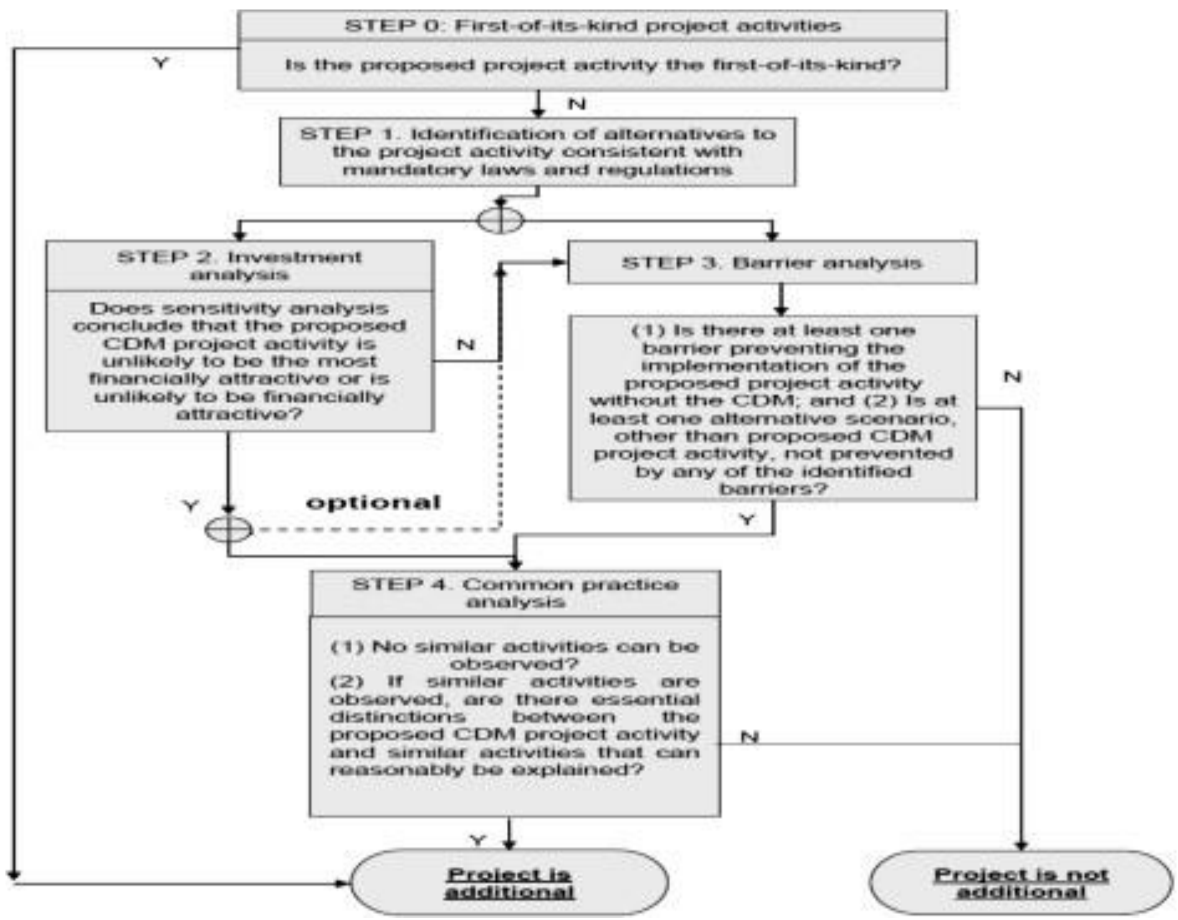
[26] Electricity Act. 2003 and its subsequent amendments

[27] https://www.iitr.ac.in/wfw/web_ua_water_for_welfare/power/Electricity_Act_2007.pdf

The both alternatives (as discussed below) are in compliance with laws and regulations required. There is no any mandatory requirement to implement the project activity.

Additionality Assessment for large scale project activity Instances

The present project generates power using Solar PV energy which is a renewable, zero emission source of energy. Baseline considerations for the project are based on approved consolidated baseline methodology AM0123 (Version 1.0). The methodology requires the project investor to determine the additionality based on "Methodological Tool- Tool for the demonstration and assessment of additionality", Version 7.0.0. The step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs.



The additionality of the Project activity is ascertained in line with the applicable guidance from the UNFCCC. The demonstration of additionality for the proposed Project activity is being carried out in accordance with the additionality tool provided by the UNFCCC i.e., "Tool for demonstration and assessment of Additionality" Version 7.0.0, the tool provides a step-wise approach to demonstrate additionality which is displayed below:

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

The project activity is a large-scale solar power project undertaken in Thoothukudi district, Tamil Nadu, India. This is not the first such project to be installed in the country or in the state and therefore project activity does not meet this criterion.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity: Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed project activity.

The project activity is aimed to generate power from the solar power plant using renewable energy (solar) resources and selling to identified user where as in the baseline those were using grid. Hence, the following alternatives are considered:

Alternative 1: The proposed project activity undertaken without being registered as a Proposed project activity.

The option of project activity being undertaken without being registered as a project activity is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the Fertilizer plant through India's National Grid, which has a number of power plants including existing and upcoming ones which would be used in the absence of project activity. In the absence of project activity, power from the same grid will continue to provide for the requirement of Plant and the country.

Alternative 2: Continuation of the current situation (No proposed project activity and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid).

In such case the project proponent would have continued without investment in project activity and continued with its usual business activities. In such case grid would continue with the fossil fuel-based power projects and this would result in GHG emissions. Hence, existing power plant and the new capacity add-on from a fossil fuel-based power plant is appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1a: Identified realistic and credible alternative scenario (s) to the project activity.

Of the two alternatives outlined above, the first alternative is not possible as project activity is not viable without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology. Therefore, continuation of the current situation is the likely alternative to the project activity.

The project being a green field project activity the baseline scenario in line with the methodology is “the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system Version 7.0”

Sub-step 1b: Consistency with mandatory laws and regulations: The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution.

Installation of a large-scale solar power project including the one as project activity is consistent with mandatory laws and regulations at national (India) and sub-national level (state of Tamil Nadu). The environmental regulations, legislations and policy guidelines in respect to the project activity is governed by Ministry of Environment, Forest and Climate Change (MoEF&CC), Delhi supported by Central Pollution Control Board (CPCB). In accordance to Annexure-II MOEF&CC, OM on J-11013/41/2006-IA. II (I) dated on 07-july-2017 Solar Photovoltaic Power Projects are not covered under the ambit of EIA Notification, 2006 and its subsequent amendments and hence, it does not require preparation of EIA and pursuing Environmental Clearance from MoEF&CC.

Moreover, Solar Power Project is considered under “white category” for Consent to Establish/Operate as per regulation, and thus there is no necessity of obtaining the Consent to Operate’’. Since project activity falls under white category and the non-polluting in nature, the project fulfils the compliance to the local environmental laws and regulations.

Overall, the project activity is consistent with the following mandatory laws and regulations.

- Electricity Act, 2003 and it’s subsequent amendment as it does not influence the choice of fuel used for power generation nor has it mandated power generation using renewable energy only.
- Environmental (Protection) Act, 1986 and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006 and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003.
- Noise Pollution (Regulation and Control) Amendment Rules, 2017
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016 and amended in November, 2021.

- E-waste (Management and Handling) Rules, 2020.
- Bio-Medical Waste (Management and Handling) Rules 2016
- Plastics Waste Management Rules, 2016 amended in 2021.
- Batteries (management and handling) rules 2019

Outcome of Sub-step 1b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations

Hence, both the alternatives enlisted above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations. However, Alternative 2 "Continuation of the current situation" has been selected as the appropriate baseline alternative for this project activity in line with methodology.

Step 2: Investment analysis

Investment analysis is carried out to determine on whether the proposed project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of emission reductions credits. This is demonstrated in line with the Tools for sections as per "Investment Analysis" (Ver 12.0)

Sub-step 2a: Determine appropriate analysis method

There are three options for the determination of analysis method which are:

I. Simple Cost Analysis

II. Investment Comparison Analysis and

III. Benchmark Analysis

The Project activity envisages to export the power to Indian grid and the revenues from the sale of electricity would be generated in accordance with the terms and tariffs established in the Power Purchase Agreement (PPA) signed between Greenam Solar and Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL). Thus, simple cost analysis (Option I) cannot be used as the analysis method as the sale of the units of generated electricity shall result in a revenue stream during the operations of the Project activity

In the absence of the project activity grid electricity would have been the obvious choice for the Project which requires no investment. Hence investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option, I and Option II, the use of Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be employed for analysis of the said project.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen. Further, this method illustrates that the evaluation of the project by the project proponent was carried out before the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

According to the “Tool for demonstration and assessment of Additionality”, Version 07.0.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax equity IRR.

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the Appendix, i.e., Default values for cost of equity (expected return on equity) is presented below:

Appendix of the CDM TOOL 27: Investment Analysis Version 12[28].0 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in India = 9.77%

The Required return on equity (benchmark) was computed in the following manner:

Nominal Benchmark = {(1+Real Benchmark) *(1+Inflation rate)}-1

Where:

Default value for Real Benchmark = 9.77% (as per Appendix of CDM TOOL 27: Investment Analysis)

Inflation Rate forecast for by Reserve Bank of India (RBI) (i.e. Central Bank of India) for India.

Benchmark estimation:

Project	**Date of Investment Decision (Board meeting date)**	**Commissioning date**
Floating Solar project of Greenam Energy	23-January-2018	12-January-2022

Inflation Rate (at the time of investment decision making)

[28] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf

Inflation forecast (WPI Mean of annual forecasts of Non-food Manufactured Products for 10 years)	Document Date	Sources Link
4.%	06-April-2017	(Table. A.9 – Annual average percentage change) https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=17458

Benchmark for Floating Solar project of Greenam Energy

Description	Value	Sources Link
Default Value for India as per UNFCCC guidelines	9.77%	EB 116, Annex 2, TOOL27 "Investment Analysis", Version 12.0 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf
Inflation forecast (WPI Mean of annual forecasts of Non-food Manufactured Products)	4.%	(Table. A.9 – Annual average percentage change) https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=17458
Benchmark	14.16%	Calculated

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III): The Post tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e. 25 years. It is calculated based on the cash outflows from and cash inflows into the project activity. substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

The input assumption referred below:

Technical Parameters			
Particulars	**Value**	**Unit**	**Source**
Capacity of the plant	22	MW	DPR
Plant Load factor	21.85%	%	DPR
Grid Availability	100	%	As a conservative approach
Auxiliary Consumption	0.4%	%	DPR
Transmission Loss & Charges (Wheeling charges)	3.91%	%	as per TNERC Order TP No:2 of 2017 dt 11.08.2017
Annual Module degradation	0.43%	%	DPR

Working days in a year	365	Days	DPR
Working hours in a day	24	Hrs	DPR
Technical Lifetime of the project	25	Years	DPR

Financial Parameters			
Particulars	**Value**	**Unit**	
Annual O & M cost	7.4	Million INR/annum	DPR, page 84
Escalation in O&M	5.00%	%	DPR
Insurance	0.09%	%	DPR
Admin Expenses	0.14	Million INR/annum	Opinion of sector expert (Salary etc)
Escalation in Admin Expenses from 2nd year onwards	5.00%	%	Opinion of sector expert (Salary etc)
Inverter warranty	0.78	Million	Opinion of sector expert
Project cost	1274	Million INR	DPR, page 87
Equity @ 25% of total cost	318.5	Million INR	DPR, page 87
Debt value@ 75% of total cost	955.50	Million INR	DPR, page 87
Residual/scrap/salvage Value	10%	%	
Power Tariff	4.12	INR/ kWh	DPR
Depreciation (Book depreciation) on civil works	10%	%	https://taxadda.com/depreciation-rates-as-per-companies-act-2013/
Depreciation (Book depreciation) on plant & machineries	8%	%	https://taxadda.com/depreciation-rates-as-per-companies-act-2013/

IT depreciation on building & civil works	7.69%	%	IT depreciation rate[29]
IT depreciation on Plant & Machineries	40%	%	IT depreciation rate[30]
Corporate Tax Rate	25.17%	%	https://www.pwc.com/mu/en/services/india-desk/corporate-tax.html
Interest rate on loan	10.66%	%	https://cercind.gov.in/2017/orders/05.pdf (page 5)
Commercial operation date	12-January-2022		Commissioning certificate

Outcome of Step 2:

Calculation of equity IRR is submitted in the form of excel sheet.

Equity IRR	Benchmark (Equity IRR)
5.17%	14.16%

This substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark Equity IRR). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

Sub-step 2d: Sensitivity Analysis

Addressing Guidance under Investment Analysis Version-12.0, following factors has been subjected to sensitivity analysis:

1. PLF
2. O&M Cost
3. Project Cost
4. Tariff

The rationale of sensitivity is, "The ultimate objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the suitability of the assumptions used in the development of the investment analysis."

[29] https://search.incometaxindia.gov.in/pages/results.aspx?k=7%2E69

[30] https://search.incometaxindia.gov.in/pages/results.aspx?k=7%2E69

Sensitivity Analysis

The results of sensitivity analysis show that the even with a variation of +10% & -10% in the project cost, O&M cost, PLF and Tariff Rate Equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favourable conditions.

Probability to breach the benchmark:
Sensitivity Parameter 1 : PLF or generation
PLF considered in financials for is as per Third Party report in line with "Guidelines for the reporting and validation of Plant load factors" stated in EB 48 Annex11 option 3(b). Hence, variation in PLF of more than 10% is unlikely to happen as the PLF has been reported as per the Third Party Report based on long term data. IRR is not crossing the benchmark if PLF increases by 10% in any circumstances.
Sensitivity Parameter 2: O&M
The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is generally subject to escalation and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. Hence, the reduction in the O&M cost is highly unlikely.
Sensitivity Parameter 3 : Project Cost
Project Cost for financial analysis is considered based on the information of DPR, being available at the time of investment decision making to go ahead with the project activity. However, the Sensitivity is carried out for 10% variation and for threshold level below which benchmark is not breached. Thus, it is unlikely that project cost will change beyond sensitivity range
Sensitivity Parameter 4 : Tariff Rate
The tariff is fixed for entire lifetime of the project activity. Hence, there is no probability to get variation for the same. However, Sensitivity is carried out for +/-10% even then the benchmark is not breached.

Result of Sensitivity Analysis Table:

Parameters	-10%	IRR (Base Case)	10%
PLF or Generation	1.33%	5.17%	9.01%
O&M cost	5.46%	5.17%	4.87%
Project Cost	8.98%	5.17%	2.12%
Tariff	1.33%	5.17%	9.01%

Demonstration of Additionality with Actual IRR

Since the project activity is already commissioned and operating for more than one year. Project activity IRR is calculated based on actual PLF, project cost, O&M and Tariff. Actual IRR with main four input parameters is coming -1.95% which less than the benchmark, hence project activity is deemed additional in actual scenario.

Step 3: Barrier Analysis

As per Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 or Step 3 or both can be used to demonstrate additionality of the project activity. In this case, Step 3 is not being used for the purpose.

Step 4: Common Practice Analysis

As per para 57 of Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 analysis shall be complemented with an analysis of extent to which the proposed project type (e.g., technology or practice) has already diffused in the relevant sector and region. This test is a credibility check to complement the investment analysis.

Common Practice Analysis – 22 MW Floating PV Solar Power Project at Tamil Nadu by Greenam Energy pvt Ltd

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

Range	Capacity	Unit
+50%	33	MW
Capacity of the project activity	22	MW
-50%	11	MW

Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- The projects are located in the applicable geographical area;
- The projects apply the same measure as the proposed project activity;
- The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
- The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the proposed project plant;
- The capacity or output of the projects is within the applicable capacity range for the chosen projects.
- The projects started commercial operation before the PD is published for global stakeholder consultation or before the start date of project activity, whichever is earlier for the project activity.

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- As the project is located in Tamil Nadu state of India, therefore, projects in the geographical area Tamil Nadu have been chosen for analysis. The project activity involves generation of electricity from solar energy. The project activity is located in the states of Tamil Nadu in India and the policy applicable for the solar projects is regulated by respective state policy. The policies/tariff for each state is regulated by State Electricity Regulatory Commissions of respective states and they differ for respective states. The project is Floating PV solar and a 30% equity-based project and generated power will be used by identified user through grid and paying wheeling charges. Hence, all state electricity regulation for such projects only applicable for this project. So, solar projects, which are exporting electricity to grid are followed different state electricity regulatory policy.

- The project activity is a green-field solar power project and uses measure (b) "Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies". Therefore, projects applying same measure (b) are candidates for similar projects.
- The energy source used by the project activity is solar. Hence, only solar energy projects have been considered for analysis.
- The project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.
- The capacity range of the projects is within the applicable capacity range from 11 MW to 33 MW.
- Purchase Order date of the project activity is 28-August-2018, therefore, projects, which have started commercial operation before 28-August-2018, have been considered for analysis.

Numbers of similar projects identified, which fulfil above-mentioned conditioned are N solar = 0

Step (3): Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number, Nall.

Project activities, which have got registered or are under validation with CDM/VCS/GS/GCC have been excluded in this step. The list of the power plants identified is provided to the VVB. After excluding the registered and under validation projects the total number of projects.

Nall = 0[31]

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number Ndiff.

As per the tool on Common Practice, the project activities have been separated from the different technologies on the basis two criteria:

Different technologies - are technologies that deliver the same output and differ by at least one of the following (as appropriate in the context of the measure applied in the proposed project activity and applicable geographical area):

[31] CEA data base No/CEA/RPM/09/03/2019/1470 date 20/03/2020 https://cea.nic.in/wp-content/uploads/2020/04/Plant-wise-details-of-RE-Installed-Capacity-merged.pdf

(a) Energy source/fuel (example: energy generation by different energy sources such as wind and hydro and different types of fuels such as biomass and natural gas);

(b) Feed stock (example: production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of alternative fuels or less carbon-intensive fuels);

(c) Size of installation (power capacity)/energy savings:

(i) Micro (as defined in paragraph 24 of decision 2/CMP.5 and paragraph 39 of decision 3/CMP.6); (ii) Small (as defined in paragraph 28 of decision 1/CMP.2);

(iii) Large.

(d) Investment climate on the date of the investment decision, inter alia:

(i) Access to technology;

(ii) Subsidies or other financial flows;

(iii) Promotional policies;

(iv) Legal regulations.

(e) Other features, inter alia:

(i) Nature of the investment (example: unit cost of capacity or output is considered different if the costs differ by at least 20 %).

The Project is 30% equity-based and Floating PV solar grid connected project and deployed for export of power after paying wheeling charges. Hence, the investment climate of the project is different from any other project of different tariff or revenue structure. So solar power project that are not covered under reverse bid project are considered as different from the project activity.

Hence, projects where either of the conditions is satisfied those projects are counted for calculating

Ndiff projects.

N diff = 0[32]

Step (5): calculate factor F=1-Ndiff/Nall representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

The factor for the project activity is calculated as follows:

In terms of the formula: F=1-Ndiff/Nall, the result is mathematically meaningless.

[32] CEA data base No/CEA/RPM/09/03/2019/1470 date 20/03/2020 https://cea.nic.in/wp-content/uploads/2020/04/Plant-wise-details-of-RE-Installed-Capacity-merged.pdf

Nall - Ndiff=0[33]

Thus, the project does not fulfil the requirement that F is greater than 0.2 and Nall-Ndiff is greater than 3, and the project is not a common practice

3.6 Methodology Deviations

This section is not applicable, as there is no deviation of methodology applied by the Project.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

Chronology of important dates

Activity	Date
Board Resolution date	23–January-2018
Purchase order of Module and main equipment	28-August-2018
Local Stake holder meeting	11-March-2022
Commissioning of Energy meter and Project activity	12-January-2022
Listing project for global stake holder consultation	24-March-2023
Signing contract of VVB	31-January-2023

There are no such incidents in the current monitoring period that may impact the GHG emission reductions or removals and monitoring. However, there are some occasions when energy generation was stopped due to breakdowns. List of breakdowns are attached below'

no	Date	Breakdown detail	Generation Loss (in Units)	Remarks
		FY 2021-22		
		NIL		
		FY 2022-23		
1	15-April-2022	IDT -1 Transformer got tripped in Winding temp high.	6,000	IDT -1 taken in line after adjusting the set values

[33] CEA data base No/CEA/RPM/09/03/2019/1470 date 20/03/2020 https://cea.nic.in/wp-content/uploads/2020/04/Plant-wise-details-of-RE-Installed-Capacity-merged.pdf

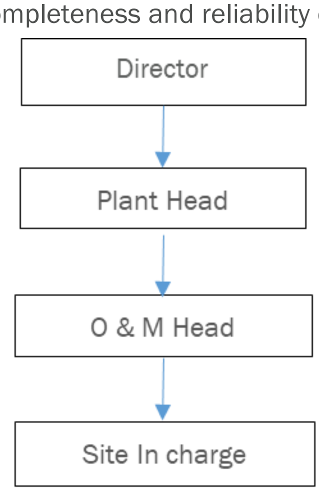
		Inverter 1 & 2 Generation affected for 1.15 hrs		in discussion with Manufacturer.
2	21-Febraury-2023	Greenam 110 KV feeder trip due to distance protection. Plant generation affected for 10.00 hrs. From 06:30 hrs to 16:30 hrs	94,000	110 KV tower line inspected and tested by TNEB(State Power utility company) and power supply resumed.
		Total Losses	100,000	
		FY 2023 -24		
1	12-May-2023	Greenam 110 KV feeder trip due to distance protection. Plant generation affected for 7.00 hrs. From 06:15 hrs to 13:15 hrs	69,000	110 KV tower line inspected by TNEB (State Power Utility Company)and power supply resumed.

Organizational Structure for Monitoring

Responsibilities of Head (Director): Tracking and reviewing the overall functioning and maintenance of the project activity from Plant Head (Operations). Plant Head (Operations) will be reporting Director.

Responsibilities of Plant Head : Overall functioning of the project activity and Coordinating with the O & M Manger and Team for the proper functioning of Project activity. He will be reporting to Director.

Responsibilities of O & M Head: O & M Manager and team is responsible for Operations and Maintenance related issues, they are also responsible for day-to-day data collection and monitoring, ensures completeness and reliability of data (calibration of equipment).



The Site In-charge will be responsible for daily reporting and checking of energy generation. Daily reading are verified by Plant head. All the values from generation record will be checked with JMR and invoices for consistency. In case there are any non-conformances identified. The Site In-charge will investigate the error and revise the record to correct it. In any case where values have slightest of variation in different records the most conservative value will be taken in the project monitoring report

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

As per the approved consolidated Methodology AM00123 (Version 1.0) para 39: Baseline emissions include only CO_2 emissions from electricity generation either in grid-connected and or captive fossil fuel fired power plants that are displaced due to the project activity. The baseline emissions are to be calculated as follows:

Baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid- connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{CO2,y} + EG_{PJsurplus,y} \times EF_{grid,y} \quad \text{Equation (6)}$$

Where:

BE_y	=	Baseline emissions in year *y* (t CO_2/yr)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and supplied to a captive consumer as a result of the implementation of the CDM project activity in year *y* (MWh/yr)
$EG_{PJsurplus,y}$	=	Quantity of the surplus electricity supplied to the grid as a result of the implementation of the CDM project activity in year *y* (MWh/yr)
$EF_{CO2,y}$	=	The electricity emission factor (t CO_2/MWh) of the baseline electricity source
$EF_{grid,y}$	=	The emission factor of the grid (t CO_2/MWh) following the procedures described in TOOL07

Electricity from Grid is baseline in this project activity hence EF_{CO2} is equivalent to $EF_{grid,y}$

The calculation of $EG_{PJ,y}$ for net generation is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y} \quad \text{Equation (7)}$$

Where:

$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and supplied to a captive consumer as a result of the implementation of the CDM project activity in year *y* (MWh/yr)
$EG_{facility,y}$	=	Quantity of net electricity generation supplied by the project plant/unit to the captive consumer in year *y* (MWh/yr)

As per methodology, combined grid emission factor as per the “Tool to calculate the emission factor for an electricity system” version 7.0 is calculated as below:

Annual Baseline emission for first year of crediting period. There will be yearly reduction in energy generation due to annual module degradation. Project activity has considered 0.43% as annual module degradation

Capacity (MW)	PLF (%)	Auxiliary Consumption	Transmission loss	Net Generation	Baseline emissions factor	Baseline emissions	Emission reductions
MW	%	%	%	(MWh/year)	(tCO_2/MWh)	(tCO_2/year)	(tCO_2/year)
22	21.85	0.4	0.391	41,776	0.9310	38,893	38,893

As per the methodology, combined grid emission factor is estimated as per the “Tool to calculate the emission factor for an electricity system” version 07, combined margin is calculated as below:

CO_2 Baseline Database for The Indian Power Sector, Version 18, September 2022 published by Central Electricity Authority (CEA), Government of India has been used for the calculation of emission reduction.

Step 1: Identify the relevant electricity systems:

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional):

Step 3: Select a method to determine the operating margin (OM):

Step 4: Calculate the operating margin emission factor according to the selected method:

Step 5: Calculate the build margin (BM) emission factor:

Step 6: Calculate the combined margin (CM) emission factor.

Step1: Identify the relevant electricity systems

As described in tool "For determining the electricity emission factors, identify the relevant project electricity system. Similarly, identify any connected electricity systems". It also states that "If the DNA of the host country has published a delineation of the project electricity system and connected electricity systems, these delineations should be used". Keeping this into consideration, the Central Electricity Authority (CEA), Government of India have Indian grid. However, since August 2006, however, all regional grids except the southern Grid had been integrated and were operating in synchronous mode, i.e., at same frequency. Consequently, the Northern, Eastern, Western and North-Eastern grids were treated as a single grid named as NEWNE grid from FY 2007- 08 onwards for the purpose of this CO2 Baseline Database. As of 31 December 2013, the southern grid has also been synchronized with the NEWNE grid, hence forming one unified Indian Grid. Since the project supplies electricity to the Indian grid, emissions generated due to the electricity generated by the Indian grid as per CM calculations will serve as the baseline for this project.

Table: Geographical Scope of Indian Electricity Grid

Northern	Eastern	Western	North-Eastern	Southern
Chandigarh	Bihar	Chhattisgarh	Arunachal	Andhra
Delhi	Jharkhand	Gujarat	Assam	Karnataka
Haryana	West Bengal	Daman & Diu	Manipur	Kerala
Himachal Pradesh	Sikkim	Dadar & Nagar Haveli	Meghalaya	Tamil Nadu
Jammu & Kashmir	Andaman &Nicobar	Madhya Pradesh	Mizoram	Telangana
Punjab		Maharashtra	Nagaland	Puducherry
Rajasthan		Goa	Tripura	Lakshadweep
Uttar Pradesh				
Uttarakhand				

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation. The Project Participants has chosen only grid power plants in the calculation.

Step-3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor (EFgrid,OM,Y) is based on one of the following methods, which are described under Step 4:

a) Simple OM: or

b) Simple adjusted OM: or

c) Dispatch data analysis OM: or

d) Average OM.

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project developers. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/ must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, solar, low-cost biomass, nuclear and solar generation.

Share of Must-Run (Hydro/Nuclear) (% of Net generation)

	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22
India	14.6%	14.3%	14.5%	17.0%	16.5%	15.8%

Data Source: Central Electricity Authority (CEA) database Version 18.0, October 2021[34]

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the Indian grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation

The Simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (t CO_2/MWh) of all generating power plants serving this system, not including low-cost/must-run power plants/units.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the following data vintages.

(**a) Ex-ante options**: If the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emission factor during the crediting period is required

[34] https://cea.nic.in/cdm-co2-baseline-database/?lang=en

b) Ex-post option: If the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

Project Proponent has chosen ex-ante option for calculation of Simple OM emission factor using a 3-year generation-weighted average, based on the most recent data available at the time of submission of PSF for validation.

OM determined at validation stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

Step 4: Calculate the operating margin emission factor ($EF_{grid,OMSimple,y}$) according to the selected method

The operating margin emission factor has been calculated using a 3-year data vintage:

Net Generation in Operating Margin (GWh) (incl. Imports)				
		2019-20	2020-21	2021-22
INDIAN Grid		965,009	958,218	1,035,672

Simple Operating Margin (tCO_2/MWh) (incl. Imports)				
		2019-20	2020-21	2021-22
INDIAN Grid		0.9541	0.9402	0.9605

Weighted Generation Operating Margin	
INDIAN Grid	**0.9518**

Step 5: Calculate the build margin (BM) emission factor ($EF_{grid,BM,Y}$)

As per Methodological tool" Tool to calculate the emission factor for an electricity system" (Version 07.0, EB 100, Annex 4) para 72:

In terms of vintage of data, project participants can choose between one of the following two options:

a) Option 1- for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group at the time of PDMR submission to the VVB. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the VVB. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

b) Option 2- For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex-ante as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

Option 1 as described above is chosen by Project Proponent to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of PSF and is fixed for the entire crediting period.

Option 1 as described above is chosen by Project Proponent to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of PSF and is fixed for the entire crediting period

Build Margin (tCO_2/MWh) (not adjusted for imports)	
	2021-22
INDIAN Grid	0.8687

Step 6: Calculate the combined margin (CM) emission factor ($EF_{grid,CM,y}$)

As per Methodological too "Tool to calculate the emission factor for an electricity system" (Version 07.0, EB 100, Annex 4) para 81:

The calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM: or

(b) Simplified CM.

Project Participant has chosen option (a) i.e., weighted average CM to calculate the combined margin emission factor for the project activity

The combined margin emissions factor is calculated as follows:

$EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$

Where,

$EF_{grid,BM}$	Build margin CO_2 emission factor in year y ($t CO_2$/ MWh)

$EF_{grid,OM,Y}$	Operating margin CO_2 emission factor in year y (t CO_2/MWh)
W_{OM}	Weighting of operating margin emissions factor (per cent)
W_{BM}	Weighting of build margin emissions factor (per cent)

The following default values should be used for WOM and WBM:

For solar project activities W_{OM} =0.75 and W_{BM} = 0.25 (owing to their intermittent and non-dispatchable nature) for the second crediting period and for subsequent crediting periods. Since project activity is of power generation by using solar, the above weightage has been considered for OM and BM.

Therefore, $EF_{grid,CM.y}$ =0.9518*0.75+ 0.8687*0.25

= 0.9310 CO_2/MWh

Baseline emission factor (EF_Y):

The baseline emission factor is calculated using the combined margin approach as described in Step 6 above:

Therefore, $EF_y = EF_{grid,CM,y}$ = 0.9310 tCO_2/MWh

5.2 Project Emissions

As per para 25 of the applied methodology AM0123 (version 01.0), no project emissions are considered in this project activity.

Hence, project emissions PEy = 0 tCO_2e

5.3 Leakage

As per para 42 of the methodology AM0123 version 01.0, Leakage emission as per methodology is "The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected".

5.4 Estimated Net GHG Emission Reductions and Removals

The annual emission reductions ERy for the project activity are calculated as the baseline emissions minus the project emissions. Hence the Project emission and Leakage emission is 0. Therefore, $ER_Y = BE_Y$

Yearly emission reduction projection:

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
Year 1 (12-January-2022 to 11-January-2023)	38,893	0	0	38,893
Year 2 (12-January-2023 to 11-January-2024)	38,726	0	0	38,726
Year 3 (12-January-2024 to 11-January-2025)	38,559	0	0	38,559
Year 4 (12-January-2025 to 11-January-2026)	38,394	0	0	38,394
Year 5 (12-January-2026 to 11-January-2027)	38,229	0	0	38,229
Year 6 (12-January-2027 to 11-January-2028)	38,064	0	0	38,064
Year 7 (12-January-2028 to 11-January-2029)	37,900	0	0	37,900
Total	268,765	0	0	268,765

However vintage wise ER calculation are as per below table

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
Year 1 (12-January-2022 to 31-December-2022)	37,614	0	0	37,614
Year 2 (01-January-2023 to 31-December-2023)	38,726	0	0	38,726
Year 3 (01-January-2024 to 31-December-2024)	38,559	0	0	38,559

Year 4 (01-January-2025 to 31-December-2025)	38,394	0	0	38,394
Year 5 (01-January-2026 to 31-December-2026)	38,229	0	0	38,229
Year 6 (01-January-2027 to 31-December-2027)	38,064	0	0	38,064
Year 7.a from (01-January-2028 to 31-December-2028)	37,900	0	0	37,900
Year 7.b from (01-January-2029 to 11-January-2029)	1,142	0	0	1,142
Total	268,629	0	0	268,629

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid, OM, y}$
Data unit	tCO_2/MWh
Description	Operating Margin CO_2 emission factor in the year y
Source of data	CO_2 Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.9518
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	For baseline emissions calculation

Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	$EF_{grid, BM, y}$
Data unit	tCO_2/MWh
Description	Build Margin CO_2 emission factor in the year y
Source of data	CO_2 Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.8687
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	For baseline emissions calculation
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	$EF_{grid, CM, y}$
Data unit	tCO_2/MWh
Description	Combine Margin CO_2 emission factor in the year y
Source of data	CO_2 Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.9310
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	For baseline emissions calculation
Comments	This parameter is fixed ex-ante for the entire crediting period.

6.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh/year
Description	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y
Source of data	Monthly Joint Meter reading know as AMR (auto meter Reading)
Description of measurement methods and procedures applied	This Solar Plant is having 2 no.s of Tariff meters (Main & Check) in Metering point. These meters readings are automatically stored in meter (Billed) on every month 1st 00:00 hrs. This reading (4 digit accuracy) will be automatically uploaded to TNEB main server at their HQ. Based on these readings the Monthly Generation statement will be developed in 4 digit accuracy automatically and the same will be displayed in consumer's portal (Greenam OA portal) at TNEB website. Note: TNEB officials are taking the recorded meter readings (2 digit accuracy) on 1st of every month for cross checking the meter working condition.
Frequency of monitoring/recording	Continuous measurement & monthly recording
Value applied:	41,776 MWh (first year) 41,241 MWh (Average annual net generation for 7 years after considering degradation factor)
Monitoring equipment	**Main meter** Type of meter(s): E3M024, 3Ph, 4wire Location of meter(s): At plant switch yard. Accuracy of meter(s): 0.2S Serial number of meter(s): TNW02796 (Meters are subjected to replaced or changed as per requirement). Calibration frequency: Once in five years Date of Calibration/ validity: 26-*October*-2021

	Reference No. of Calibration Certificates	GEPC/2021-22/310/CC/2774
	Calibration Status	Ok
	Check meter	
	Type of meter(s)	E3M024, 3Ph, 4wire
	Location of meter(s)	At plant switch yard.
	Accuracy of meter(s)	0.2S
	Serial number of meter(s)	TNW02797 (Meters are subjected to replaced or changed as per requirement).
	Calibration frequency	Once in five years
	Date of Calibration/ validity	26-October-2021
	Reference No. of Calibration Certificates	GEPC/2021-22/310/CC/2775
	Calibration Status	Ok
QA/QC procedures applied	The meters are approved, tested & sealed by the State Utility and are in the custody of State Utility. The metering arrangement, accuracy class of meters, calibration frequency is under control of state utility and Project proponent do not have any control on it. The calibration of all the meters is planned to be is carried out in-line with the National standard which recommends at least once in 5-year calibration. Faulty meters will be duly replaced. The meters are of accuracy class 0.2s • Tariff metering point is consisting of Main & Check meters for both Import & Export. • Meter will reset automatically at 00:00 hrs on 1st of every month and it is AMR (Automatic Meter Reading) reading for both Import & Export. So apportioning is not required. • All the reading values are stored in the location of "Billing". • The stored readings are transferred to Main server from Tariff meter thro' Modem. • The stored readings will be taken by TNEB-Tangedco officials manually on 1st of every month from the meters storage location (Billing). • From the Server the AMR readings & Generation statement will be generated and will be available in the OA portal of the consumer in Tangedco website and it can be downloaded In absence or delay in the meter calibration appropriate Guidelines will be applied appropriately to confirm the conservativeness of metering. Monthly generation Export can be cross checked from Invoices.	
Purpose of data	To calculate baseline emission	
Calculation method	The Net electricity exported/supplied to the grid by the project activity is estimated as a difference of electricity exported to the grid and electricity imported from the grid.	

	The net electricity supplied/exported to the grid can be obtained from the record of Monthly Joint Meter Reading by the state utility Thus, Net electricity supplied to the grid by the project plant in a given month = Export, kWh– Import, kWh.
Comments	Data will be archived in paper & electronically for a period of 2 years beyond the end of crediting period or of the last issuance of credits for this project activity, whichever occurs later.

6.3 Monitoring Plan

The monitoring plan is developed in accordance with the modalities and procedures for CDM project activities and is proposed for grid-connected solar power project being implemented. The monitoring plan, which will be implemented by the project Proponent describes about the monitoring organisation, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

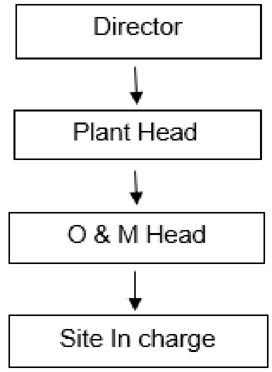
The authority and responsibility for registration, monitoring, measurement, reporting and reviewing of the data rests with the project participant. PO proposed the following structure for data monitoring, collection, data archiving and calibration of equipment's for this project activity. The team comprises of the following members:

Organizational Structure for Monitoring

Responsibilities of Head (Director): Tracking and reviewing the overall functioning and maintenance of the project activity from Plant Head (Operations). Plant Head (Operations) will be reporting Director.

Responsibilities of Plant Head: Overall functioning of the project activity and Coordinating with the O & M Manger and Team for the proper functioning of Project activity. He will be reporting to Director.

Responsibilities of O & M Head: O & M Manager and team is responsible for Operations and Maintenance related issues, they are also responsible for day-to-day data collection and monitoring, ensures completeness and reliability of data (calibration of equipment).



The Site In-charge will be responsible for carrying out internal auditing and QA/QC. All the values from generation record will be checked with JMR and invoices for consistency. In case there are any non-conformances identified. The Site In-charge will investigate the error and revise the record to correct it. In any case where values have slightest of variation in different records the most conservative value will be taken in the project monitoring report.

Data Measurement

The export and import energy will be measured continuously using Main and Check meters located at the substations. Readings of meters shall be taken on monthly basis by authorized officer of utility in the presence of the representative of Project Proponent (PO). Based on the Meter Reading Statement, invoices will be raised. These invoices can be used for cross checking the meter readings taken for the respective project activity.

Data Collection and Achieving

Readings from meters will be collected in the presence of the plant in-charge. Export and Import data would be recorded and stored in logs as well as in electronic form. The records are checked periodically by the Plant Manager and discussed thoroughly with the plant supervisor. The period of storage of the monitored data will be 2 years after the end of crediting period or till the last issuance of VCUs for the project activity whichever occurs later.

Emergency Preparedness

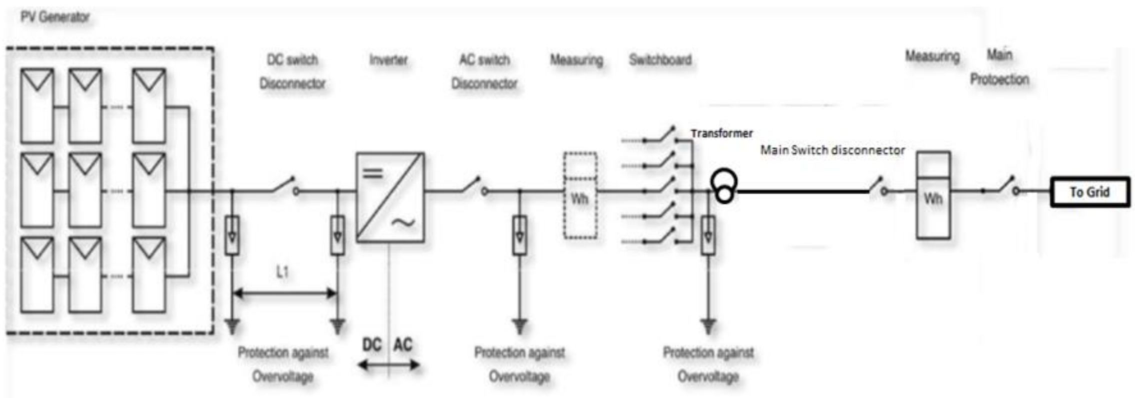
The project activity will not result in any unidentified activity that can result in substantial emissions from the project activity. No need for emergency preparedness in data monitoring is visualized. In the event that the main meter, which is used to record the net electricity exported by the project, is found to be faulty it will be repaired or replaced by authorized officer of SEB (State Electricity Board) and the data from the check meter will be used in its place. In the unlikely event that the check meter fails it will also be repaired or replaced.

Personnel Training

In order to ensure proper functioning of the project activity and proper monitoring of emission reductions, the staff will be trained. The plant staff will be trained in equipment operation, data recording, reports writing, operation & maintenance, and emergency procedures in compliance with the monitoring plan.

Metering Arrangement

Line diagram with metering arrangement for the solar project activity is shown below:



QA/QC Procedures

The energy meters at the feeders are maintained and owned by state electricity board. Neither the project proponent nor the site personnel have any control over it. The records will be crosschecked with the records of sold electricity. The meters are calibrated by state electricity board out in-line with the Nation standard which recommends at least once in 5-year calibration or whenever abnormal difference/inconsistency is observed between main meter and check meter. However in case of any findings or nonconformities identified during internal audit or during the operation the site incharge will do root cause analysis and will implement corrective and preventive action to stop future occurrences. Internal audit will be carried out once in a year.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh (Monitoring period)

Description	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y
Value applied:	38,856
Comments	-

7.2 Baseline Emissions

The baseline emissions is the product of net electricity generation supplied to the grid expressed in MWh by the project multiplied by an emission factor.

$$BE_y = EG_{PJ,y} \times EF_{CO2,y} + EG_{PJsurplus,y} \times EF_{grid,y}$$

Where:

BE_y	=	Baseline emissions in year *y* (t CO_2/yr)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and supplied to a captive consumer as a result of the implementation of the CDM project activity in year *y* (MWh/yr)
$EG_{PJsurplus,y}$	=	Quantity of the surplus electricity supplied to the grid as a result of the implementation of the CDM project activity in year *y* (MWh/yr)
$EF_{CO2,y}$	=	The electricity emission factor (t CO_2/MWh) of the baseline electricity source
$EF_{grid,y}$	=	The emission factor of the grid (t CO_2/MWh) following the procedures described in TOOL07

$BE_y = 38{,}856 \times 0.9310 + 0 \times 0.9310$ (since project activity is total captive hence Surplus electricity to grid is zero)

= 36,174 tCO_2

7.3 Project Emissions

The project is a solar farm, belongs to renewable energy activity, hence PE_y =0 tCO_2e/MWh.

7.4 Leakage

No leakage is considered for this project.

7.5 Net GHG Emission Reductions and Removals

$ER_Y = BE_Y - PE_Y$

Where:

ER_Y = Emission reductions in year y (tCO_2)

BE_Y = Baseline Emission in year y (tCO_2)

PE_Y = Project Emission in year y (tCO_2)

The total monitoring period (12-January-2022 to 28-February-2023) is of 424 days. The estimated average annual emission reduction is 38,395 tCO_2.

$ER_Y = BE_Y - PE_Y$

$ER_Y = 36{,}174 - 0$

$ER_Y = 36{,}174\ tCO_2$

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
12-January-2022 to 31 -December -2022	30,848	0	0	30,848
01-January-2023 to 28 -February -2023	5,326	0	0	5,326
Total	36,174	0	0	36,174

Ex-ante emissions reductions/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
37,219 tCO_2 (Year 2022)	30,848 tCO_2	-17.12%	The actual emission reduction is -17.12% lesser than the estimated value which is due to variation in the uncontrol climatic condition.
6,203 tCO_2 (Year 2023)	5,326 tCO_2	-14.14%	The actual emission reduction is -14.14% lesser than the estimated value which is due to variation in the uncontrol climatic condition.

APPENDIX 1: LIST OF ATTENDEES

ATTENDANCE SHEET

STAKEHOLDERS CONSULTATION MEETING

GCC Project ***: Floating Solar project of Greenam Energy***

Organizer : Greenam

Project Location : Tuticorin

VENUE: Greenam office,
SPIC Nagar

DATE: 11/03/2022

Sl. No.	Name	Designation	Signature
1.	Arumugam.M	Soosainagar,	
2.	Rajesh.P	,,	
3.	Santhana kumar	,,	
4.	Siva kumar	,,	
5.	Vijayan	,,	
6.	Jayavel	,,	
7.	Venkatesan	,,	

ATTENDANCE SHEET

STAKEHOLDERS CONSULTATION MEETING

GCC Project *: Floating Solar project of Greenam Energy*

Organizer : Greenam

Project Location : Tuticorin

VENUE: Greenam office, SPIC Nagar

DATE: 11/03/2022

8.	Ganesan	Sooraiyagar	[illegible]
9.	Rajesh	"	[illegible]
10.	Isravel	"	[illegible]
11.	Gomathinayagam	Thangamalpuram	[illegible]
12.	Suresh Kumar	"	[illegible]
13.	Thiyagarajan	"	[illegible]
14.	Melvin	"	Melvin
15.	Palavesam	"	[illegible]

ATTENDANCE SHEET

STAKEHOLDERS CONSULTATION MEETING

GCC Project ***: Floating Solar project of Greenam Energy***

Organizer : Greenam

Project Location **:** Tuticorin

VENUE: Greenam office,
SPIC Nagar

DATE: 11/03/2022

16.	Umashankar	Soosainagar	Umashankar P.
17.	Dinesh	"	[illegible]
18.	Vaanamaamalai	"	[illegible]
19.	Murugan	"	[illegible]
20.	Rajagopalan	"	[illegible]
21.	Jhonson	"	[illegible]
22.	Perumal	"	[illegible]